

Approach Document: Volleyball League Automation System

Objective:

Develop a fully open-source, role-based system for managing 80+ team volleyball tournaments. This system should automate registration, scheduling, score tracking, playoff logic, alerts, and certificate generation while being beginner-friendly, modular, and scalable. The primary platforms include Airtable (for data), Softr (for UI), Python (for logic), and n8n (for automation).

System Modules:

1. Team Registration
 2. Match Scheduler (Group Stage)
 3. Live Score Tracking
 4. Standings & Leaderboard Auto-Update
 5. Playoff Bracket Generator
 6. Notifications (WhatsApp/Email)
 7. Public Scoreboard View
 8. Certificate Generator
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Platform & Tool Stack:

- Softr: Frontend dashboards and login control
- Airtable: Data backend (Teams, Matches, Scores, Standings)
- Python: Custom logic (match scheduling, seeding, data handling)
- n8n: Workflow automation (alerts, reminders)
- GitHub: Version control and delivery repo
- Optional: Firebase for hosting frontend public scoreboard

Architecture Flow:

- Player registers via Softr → Data saved in Airtable → Admin runs Python scheduler → Results uploaded to Airtable → Coordinators input scores → Python script creates bracket → n8n sends daily alerts → Public scoreboard updated.

Data Structure:

- Teams (ID, Team Name, Email, Region)
- Matches (Date, Time, Court, Team A, Team B, Score A, Score B)
- Standings (Team ID, Points, Wins, Losses)
- Playoff Brackets (Match, Seeded Team A vs B)
- Certificates (Player, Team, Rank, URL)

Developer SOP: End-to-End Execution

Prerequisites:

- Basic knowledge of Python 3
- n8n installed locally (via Docker or desktop app)
- Airtable and Softr accounts set up
- GitHub for version tracking

Software Setup:

1. Clone the GitHub repository or extract the ZIP file into your working directory.
2. Create and activate a virtual environment:
3. `python -m venv venv`
4. `source venv/bin/activate` # For Windows: `venv\Scripts\activate`

5. Install Python dependencies:
 6. pip install pandas python-dotenv
 7. Set up .env file with necessary API credentials (optional for extensions)
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Task Plan: 3–4 Hour Work Blocks

Day 1: Local Setup and Group Match Scheduling

- Load teams.csv and validate 82 teams
- Run scheduler.py to auto-generate match_schedule.csv
- Review the generated schedule and push to Airtable manually or via Airtable CLI/API
- Create Matches table with relevant fields: Date, Time, Court, Team A, Team B
- Configure Softr login flow connected to Airtable
- Create basic player dashboard with login-to-schedule mapping

Day 2: Score & Standings + Playoff Bracket Generation

- Create Scores and Standings tables in Airtable
- Add score fields in Matches table (Score A, Score B)
- Add formula columns to calculate Points, Wins, Losses
- Export standings.csv from Airtable
- Run seeding.py → output bracket.json
- Import this bracket into a visual block on Softr

Day 3: Automation Setup (n8n) + Testing

- Install n8n locally or use n8n cloud
- Import n8n_reminder_workflow.json
- Set up Airtable API keys and base ID inside n8n Credentials
- Test scheduled trigger (daily 9 AM) to send match alerts

- Configure email/WhatsApp integration (SendGrid, WhatsApp Business API)

Day 4: Certificate System and Public Scoreboard

- Review certificate_template.html
 - Implement auto-name fill using Python or Softr button action
 - Host certificate rendering page if not embedding into Softr
 - Load audience_scoreboard.csv for public viewing
 - Configure Softr public link (read-only)
 - Final testing and documentation review
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General Practices:

- Maintain local logs (scheduler.log, seeding.log)
 - Each function in scripts should have clear docstrings
 - Daily commits to GitHub with changelogs
 - Store API keys using .env and never hard-code in scripts
 - Use Notion or README to track modules completed, in progress, or pending QA
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Final Deliverables:

- README.md (detailed guide)
- APPROACH.md (this doc)
- Python scripts with logging, docstrings, version history
- Airtable base link or template copy
- Softr UI export or screenshot deck
- n8n automation JSON
- Data files with 82 teams, matches, standings, results
- Bracket JSON + Certificate HTML templates